

SUI, XINXIN

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2275 Speedway, Austin, TX 78705

EDUCATION

Ph.D.	Civil and Environmental Engineering	University of Texas at Austin	2021 – 2025.6 (expected)
M.S.	Civil Engineering and Geosciences	Delft University of Technology, Netherlands	2017 – 2019
B.E.	Environment and Resources	Jilin University, China	2013 – 2017

EXPERIENCE

NASA FINESST Fellow	Civil and Environmental Engineering	University of Texas at Austin
● Urbanization Impacts on Extreme Precipitation ^{1,3,4,5,7}		2021 – Present
● Evapotranspiration under the Influence of Urban Morphology ²		2022 – Present
Research Assistant	Remote Sensing and Geographic Information System	Peking University, China
● Ground Validation for Satellite-Based IMERG Precipitation Dataset ⁸		2020
Research Assistant	Civil and Environmental Engineering	National University of Singapore
● Low Impact Development Influence on the Rainfall-Runoff Relationship at Catchment Scale ⁶		2019
Internship		TU Delft & Arcadis
● Sponge City Multidisciplinary Project ¹⁴		2018
Undergraduate Research		Jilin University
● Water Cycle and Utilization in Huadian City (Thesis)		2016 – 2017
● The Impacts of Oil Shale In-Situ Mining on Groundwater Temperature ¹³		2015 – 2017

RESEARCH FIELDS

Urban Hydrometeorology^{1,2,3,4,5} & Water Management^{6,14}, Data-driven approach^{1,7}, Remote Sensing^{7,8}, Hydrologic Models⁶

PUBLICATIONS

LEAD AUTHOR

- 1) **Sui, X.**, Yang, Z. & Niyogi, D. (2025). Discovering Causal Factors in Urban Climate Anomalies: Urbanization, Local Climate, or Global Warming? *In Preparation*
- 2) **Sui, X.**, Yang, Z. & Niyogi, D. (2025). Evapotranspiration under Urban Development Environments *Internal Review*
- 3) **Sui, X.**, Nielsen-Gammon, J., Yang, Z. & Niyogi, D. (2025). How Texas Cities Alter Warm-Season Storms? *Internal Review*
- 4) **Sui, X.**, Yang, Z. & Niyogi, D. (2025). Diurnal Urban Precipitation Anomalies across Diverse Landscapes. *Science Advances* *Under Review*
- 5) **Sui, X.**, Yang, Z., Shepherd, M. & Niyogi, D. (2024). Global Scale Assessment of Urban Precipitation Anomalies. *Proceedings of the National Academy of Sciences*, 121, no. 38: e2311496121.
- 6) **Sui, X.**, & van de Ven, F.H. (2023). The influence of Low Impact Development (LID) on basin runoff in a half-urbanized catchment: A case study in San Antonio, Texas. *Journal of Hydrology*, 616, p.128793.
- 7) **Sui, X.**, Li, Z., Tang G., Yang Z. & Niyogi, D. (2022). Disentangling error structures of precipitation datasets using decision trees. *Remote Sensing of Environment*, 280, p.113185. doi:10.1016/j.rse.2022.113185
- 8) **Sui, X.**, Li, Z., Ma, Z., Xu, J., Zhu, S. & Liu, H. (2020). Ground Validation and Error Sources Identification for GPM IMERG Product over the Southeast Coastal Regions of China. *Remote Sensing*, 12(24), 4154. doi:10.3390/rs12244154

CO-AUTHOR

- 9) Li, Z., Tiwari, A., **Sui, X.**, Garrison, J., Marks, F. & Niyogi, D. (2023). Studying Brown Ocean Re-Intensification of Hurricane Florence Using CYGNSS and SMAP Soil Moisture Data and a Numerical Weather Model. *Geophysical Research Letters*, 50(19), p.e2023GL105102.

- 10) Gai, Y., Long, X., Wei, Z., **Sui, X.**, Jiang, B., Chen, X., Yu, D., Qin, S. & Cui, Y. (2023). Toward a better understanding of uncertainty for satellite precipitation products over complex terrain with sparse rain gauge data. *Atmospheric Research*, 288, p.106742.
- 11) Wei, Z., Long, X., Gai, Y., Yang, Z., **Sui, X.**, Chen, X., Kan, G., Fan, W. & Cui, Y. (2022). Evaluating Remote Sensing Precipitation Products Using Double Instrumental Variable Method. *IEEE Geoscience and Remote Sensing Letters*, 19, p.1-5. doi:10.1109/LGRS.2022.3192644
- 12) Li, H., Zhong, X., Ma, Z., Tang, G., Ding, L., **Sui, X.**, Xu, J. & He, Y. (2020). Climate Changes and Their Teleconnections With ENSO Over the Last 55 Years, 1961–2015, in Floods-Dominated Basin, Jiangxi Province, China. *Earth and Space Science*, 7(3). doi:10.1029/2019ea001047
- 13) Zhang, F., **Sui, X.**, Jiang, S., & Liang, X., (2019). Experimental study on the influence of moisture content on the frozen wall in in-situ mining of oil shale. *Chemical Industry Management*, 17. (in Chinese)
- 14) Ulkü, M., **Sui, X.**, van der Lans, M., Dillon Peynado, T., Zheng, J. & Fong, C. (2018). Sponge City Project, Multidisciplinary project report, Delft University of Technology

AWARDS & SCHOLARSHIPS

● NASA FINESST Fellowship (\$150,000) – Urban Influences on Extreme Precipitation	2022 – 2025
● Recruitment Fellowship, UT Austin Graduate Program (\$36,000)	2021 – 2025
● Best PhD Talk Award (2024-2025), Department of Earth and Planetary Sciences, UT Austin	2025
● Dean's Prestigious Supplement Fellowship, UT Austin	2024
● Professional Development Award, UT Austin	2021
● Excellence Award in the Innovation & Entrepreneurship Competition (¥15,000 RMB)	2017
● Third Prize Scholarship, Jilin University	2016
● Individual's Scholarship, Jilin University	2015

MEDIA & PROFESSIONAL RECOGNITIONS

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- Featured TV and video interviews: [PBS Terra](#), [KXAN](#), “Why Are Some Cities ATTRACTING Storms While Others REPEL Them?” (2024)⁵
 - Featured press interviews: [The Washington Post](#), [Inside Climate News](#), [Science Daily](#), [Phys.org](#), etc., “Why cities are getting more rainy” (2024)⁵
 - Research findings⁵ covered by major media: [Bloomberg](#), [USA Today](#), [Yahoo! News](#), [EOS](#), etc., “Many of the World's Cities Have Gotten Wetter” (2024)⁵

CONFERENCE PRESENTATIONS

- **Sui, X.**, Yang, Z.L. & Niyogi, D., (2025). Urban Rainfall Anomalies: Regional and Diurnal Variability under Diverse Landscapes. *American Meteorological Society (AMS) Annual Meeting* (oral).
- **Sui, X.**, Leung, R., Eisma, J.A. & Tiwari, A., (2024). City and Climate Extremes: Scientific and Community Perspectives. *American Geophysical Union (AGU) Annual Meeting* (session chair).
- **Sui, X.**, Yang, Z.L. & Niyogi, D., (2024). Urban Precipitation Anomalies: Explained by Diverse Landscapes. *AGU Annual Meeting* (oral).
- **Sui, X.**, Yang, Z.L., Marshall, S., & Niyogi, D., (2024). Exploring urban precipitation anomalies: The influence of local climate and urban development. *AMS Annual Meeting* (oral).
- **Sui, X.**, Li, Z., Tang, G., Yang, Z.L. & Niyogi, D., (2023). Assessing Five Precipitation Datasets in Continental US Using Decision Tree Approach. *AMS Annual Meeting* (oral).
- **Sui, X.**, Li, Z., Tang, G., Yang, Z.L. & Niyogi, D., (2022). A novel approach to systematically evaluate precipitation datasets using decision trees. *AMS Annual Meeting* (oral).
- **Sui, X.**, Li, Z., Tang, G., Yang, Z.L. & Niyogi, D., (2022). Using Machine Learning to Assess Uncertainty in Rainfall Climatology Products. *Asia Oceania Geosciences Society Annual Meeting* (oral).
- **Sui, X.**, Li, Z., Tang, G., Yang, Z.L. & Niyogi, D., (2022). Evaluating and intercomparing satellite and reanalysis precipitation products by using decision trees. *AGU Annual Meeting* (oral).
- **Sui, X.**, Li, Z., Tang, G., Yang, Z.L. & Niyogi, D., (2021). A novel approach to systematically evaluate precipitation datasets using decision trees. *AGU Annual Meeting* (poster).

MENTORING & TEACHING

- **Anant Dimri** (UT Austin, undergraduate) – Quantifying Uncertainties in Urban Rainfall Anomalies: Insights from Multiple Precipitation Datasets 2024 – Present
- **Jingxin Liang** (UT Austin, undergraduate) – SPRI: A New Index to Quantify Drought Events 2023 – Present
- **Danny Xu** (UT Austin, undergraduate) – Assessing Freezing Rain Events: Impact of Tree Ice Loading on Austin Power Outages 2023 – 2024
- CE 398T **Supervised teaching in civil engineering** (UT Austin) – How Does the Way Students Organize Knowledge Affect Their Learning? & Academic Writing and Science Communication
- GEO 371T **Geohealth** (UT Austin) – Flood & Geohealth

SERVICE EXPERIENCES

- Invited reviewer for *Science Advances*, *Journal of Hydrology*, *Advances in Space Research*, *Journal of Remote Sensing*, *Next Sustainability*

WORKSHOP & TOWNHALL

- 2023 Planet Texas 2050 Symposium - Urban precipitation anomalies for global cities (oral). **Sui, X.**, Yang, Z.L., & Niyogi, D.
- 2023 AGU Annual Meeting - Building Your Network: Sharpening the Soft Skills of Science. Preisser, M., Jha, A., Ali, J., **Sui, X.** & Adelsperger, S., Co-Convened Town Hall
- 2023 UT Austin Portugal (Bridging the Gap: Satellite Data and Models for Local Impact | Online Workshop) - Global Urban Precipitation Anomalies Explained by Urbanization and Local Climate. **Sui, X.**, Yang, Z.L., Marshall, S., & Niyogi, D.
- 2023 AGU Hydrology Section Student Subcommittee (H3S) - Prepare for AGU Conference: Elevator Pitch Workshop. Patterson, J., **Sui, X.**, Lane, N., Becker, P., Byrnes, D.
- 2023 Cockrell School of Engineering Student DEI Board, UT Austin - DEI Talk Series (Speakers: Sripada, S., Telang, N., Castro, P., Emtiyaz, K.) Sanyal, S., Ignacio, N., **Sui, X.**, Okoronkwo, R., Goncalo, I., Udawala, S.
- 2023 Graduate Engineering Council Seminar, UT Austin - How to "Manage" Your Advisor: Professional Relationship Between Graduate Students and Faculty (Speaker: Niyogi, D.) **Sui, X.**, Ignacio, N., Maynor, K., Zhang, M., Kumar, S., Concepcion, A., Maynor, K.
- 2023 Graduate Engineering Council Seminar, UT Austin - Writing Successful Cover Letters: A Process Approach (Speaker: Spurgeon., E.). **Sui, X.**, Zhang, M., Maynor, K., Kumar, S., Maynor, K.
- 2022 Graduate Engineering Council Seminar, UT Austin - Life After Graduate School: Exploring the Options (Speaker: Lamb., J.). **Sui, X.**, Zhang, M., Maynor, K., Kumar, S., Maynor, K.
- 2018 Arcadis-Wuhan Planning & Design Institute - Urban water & transportation management workshop

VOLUNTEERING ROLES

- AGU Hydrology Section Student Subcommittee 2022 – 2024
- UT Austin Graduate Engineering Council, Seminar Chair 2022 – 2023
- UT Austin Cockrell School of Engineering Student DEI Board 2023
- Jilin University, Association of University Clubs, Vice President 2015 – 2016
- Jilin University, Student Union of the College, Vice Minister 2014 – 2015